AI Sandbox Report - Benchmark: House mini

Experiment Report: Benchmark: House mini

Experiment ID: `43859e71-a22a-404b-818a-73409d0df6c8`
Generated: 2026-04-11 13:19:50 UTC

Executive Summary

Status: ✅ Completed **Task Type:** Regression **Models Evaluated:** 10 (10 passed, 0 failed constraints) **Duration:** 14.3 seconds

Best Model: LightGBM Regressor **Best Score:** 0.3013

Recommendation

Based on weighted evaluation across performance, robustness, latency, and cost, **LightGBM Regressor** is recommended with a global score of **0.3013**. **Runner-up:** XGBoost Regressor (score: 0.2972)

Model Comparison

Rank	Model	Global Score	RMSE	MAE	R2	MAPE	Status
1	LightGBM Regressor	0.3013	180648.7890	94785.7513	0.7533	-	✅
2	XGBoost Regressor	0.2972	184342.0267	95275.4519	0.7431	-	✅
3	Random Forest Regressor	0.2968	184749.4001	97528.5303	0.7420	-	✅
4	Gradient Boosting Regressor	0.2832	196567.4166	115766.1904	0.7079	-	✅
5	Lasso Regression	0.1996	257459.1383	169357.0070	0.4989	-	✅
6	Ridge Regression	0.1996	257459.1932	169356.9866	0.4989	-	✅
7	ElasticNet	0.1982	258337.5929	169747.5888	0.4955	-	✅
8	Decision Tree Regressor	0.1902	263411.4104	135705.3302	0.4755	-	✅
9	K-Nearest Neighbors Regressor	0.1848	266745.4132	169487.9539	0.4621	-	✅
10	SVM Regressor	-0.0252	374983.3952	222213.5346	-0.0630	-	✅

Model Details

LightGBM Regressor

Rank: #1 | **Global Score:** 0.3013

All Metrics:

Metric	Value
mae	94785.7513
mse	32633984958.1984
r2	0.7533
rmse	180648.7890
test_mae	99313.1725
test_mse	39966825246.2047
test_r2	0.7445
test_rmse	199917.0459

XGBoost Regressor

Rank: #2 | **Global Score:** 0.2972

All Metrics:

Metric	Value

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r2	0.7533
rmse	180648.7890
test_mae	99313.1725
test_mse	39966825246.2047
test_r2	0.7445
test_rmse	199917.0459

XGBoost Regressor

Rank: #2 | **Global Score:** 0.2972

All Metrics:

Metric	Value
mae	95275.4519
mse	33981982809.2565

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